Recall the expansion and observe the coefficients of each term of the identities $(a+b)^0$, $(a+b)^1$, $(a+b)^2$, $(a+b)^3$. There is a pattern in the arrangements of coefficents

$$(a+b)^0 = 1 \qquad\qquad 1$$

$$(a+b)^1 = a+b \qquad\qquad 1 \quad 1$$

$$(a+b)^2 = a^2+2ab+b^2 \qquad\qquad 1 \quad 2 \quad 1$$

$$(a+b)^3 = a^3+3a^2b+3ab^2+b^3 \qquad 1 \quad 3 \quad 3 \quad 1$$

If we observe carefully the Pascal triangle, we may notice that each row starts and ends with 1 and other entries are the sum of the two numbers just above it. For example '3' is the sum of 1 and 2 above it; '10' is the sum of 4 and 6 above it. We will prove in a short while that

$$(a+b)^n = {}^n\mathrm{C}_0\, a^n b^0 + {}^n\mathrm{C}_1\, a^{n-1}b^1 + \cdots + {}^n\mathrm{C}_r\, a^{n-r}b^r + \cdots + {}^n\mathrm{C}_n\, a^0 b^n.$$

which is the ***binomial expansion*** of $(a+b)^n$. The binomial expansion of $(a+b)^n$ for any $n \in \mathbb{N}$ can be written using Pascal triangle. For example, from the fifth row we can write down the expansion of $(a+b)^4$ and from the sixth row we can write down the expansion of $(a+b)^5$ and so on. We know the terms (without coefficients) of $(a+b)^5$ are

$$a^5,\ a^4b,\ a^3b^2,\ a^2b^3,\ ab^4,\ b^5$$

and the sixth row of the Pascal triangle is

$$1 \quad 5 \quad 10 \quad 10 \quad 5 \quad 1.$$

Using these two we can write

$$(a+b)^5 = a^5+5a^4b+10a^3b^2+10a^2b^3+5ab^4+b^5.$$

The Pascal triangle can be constructed using addition alone, without using any multiplication or division. So without multiplication we can write down the binomial expansion for $(a+b)^n$ for any $n \in \mathbb{N}$.

The above pattern resembling a triangle, is credited in the name of the seventeenth century French Mathematician Blaise Pascal, who studied mathematical properties of this structure and used this concept effectively in Probability Theory.

5.2.2 Binomial theorem for positive integral index

Now we prove the most celebrated theorem called Binomial Theorem.

Theorem 5.1 (Binomial theorem for positive integral index): If n is any positive integer, then

$$(a+b)^n = {}^n\mathrm{C}_0\, a^n b^0 + {}^n\mathrm{C}_1\, a^{n-1}b^1 + \cdots + {}^n\mathrm{C}_r\, a^{n-r}b^r + \cdots + {}^n\mathrm{C}_n\, a^0 b^n.$$

Proof. We prove the theorem by using mathematical induction. For any positive integer n, let $P(n)$ be the statement

$$(a+b)^n = {}^n\mathrm{C}_0\, a^n b^0 + {}^n\mathrm{C}_1\, a^{n-1}b^1 + \cdots + {}^n\mathrm{C}_r\, a^{n-r}b^r + \cdots + {}^n\mathrm{C}_n\, a^0 b^n.$$

Since

$${}^1\mathrm{C}_0 = 1 \text{ and } {}^1\mathrm{C}_1 = 1,$$